EE2080 : Lecture 5 : 11 August 2025

PLAN: – Stopping Time w.r.t Filtrations

Stopping Time

Fix a measurable space $(\Omega, \mathcal{F})$

Let $X : \Omega \to \mathbb{R}$ be a random variable w.r.t $\mathcal{F}$

Defn: The σ-algebra generated by X, denoted $\sigma(X)$ is defined as

$$\sigma(X) = \{ A \in \mathcal{F} : A = X^{-1}(B) \text{ for some } B \in \mathcal{B}(\mathbb{R}) \}$$

Example: $\Omega = \{1, 2, \dots, 6\}$ $\quad \mathcal{F} = 2^{\Omega}$

$$X(\omega) = \begin{cases} 0, & \omega = 1, 2 \\ 4, & \omega = 5, 6 \\ 6, & \omega = 3, 4 \end{cases}$$

$$X^{-1}([-\infty, x]) = \begin{cases} \phi, & x < 0 \\ \{1,2\}, & 0 \le x < 4 \\ \{1,2,5,6\}, & 4 \le x < 6 \\ \Omega, & x \geq 6 \end{cases}$$

$$\sigma(X) = \{ \phi, \{1,2\}, \{3,4\}, \{5,6\}, \{1,2,3,4\}, \{1,2,5,6\}, \{3,4,5,6\}, \Omega \}$$

Remark: $\sigma(X)$ is the smallest σ-algebra w.r.t which X is a R.V.

σ-Algebra generated by a random vector : Let $(X_1, \dots, X_n) : \Omega \to \mathbb{R}^n$ be a random vector w.r.t $\mathcal{F}$.

$$\sigma(X_1, \dots, X_n) = \{ A \in \mathcal{F} : A = (X_1, \dots, X_n)^{-1} B \text{ for some } B \in \mathcal{B}(\mathbb{R}^n) \}$$

Filtrations : Fix a prob. space $(\Omega, \mathcal{F}, \mathbb{P})$

Let $\mathcal{T}$ be an ordered index set

Consider a collection of σ-algebras $\mathcal{G}_\cdot = \{ \mathcal{G}_t : t \in \mathcal{T} \}$ s.t $\mathcal{G}_t \subseteq \mathcal{F} \ \forall t$.

The above collection is called a filtration if $\mathcal{G}_s \subseteq \mathcal{G}_t \quad \forall s \le t$

<u>Example:</u> Let $\{X_t : t \in \mathcal{T}\}$ be a random process defined w.r.t $\mathcal{F}$, then $\mathcal{G}_t = \sigma(X_s : s \leq t)$ is called the natural filtration associated with the process $\{X_t : t \in \mathcal{T}\}$.

<u>Stopping Time:</u>

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Let $\mathcal{T}$ be an ordered index set.

Fix a filtration $\mathcal{G}_\cdot = \{\mathcal{G}_t : t \in \mathcal{T}\}$.

<u>Defn:</u> A random variable $T : \Omega \longrightarrow \mathbb{R} \cup \{\pm\infty\}$ is called a stopping time w.r.t the filtration $\mathcal{G}_\cdot$ if

- $\mathbb{P}(T < +\infty) = 1$
- For each $t \in \mathcal{T}$, $\{T \leq t\} \in \mathcal{G}_t$.

That is, the answer to the question "is $T \leq t$?" can be answered by looking at the process upto (and including) time t.